AP PRECALCULUS

Unit 1 • Worksheet 1.5C

Worksheet 1.5C • Polynomial Functions and Complex Zeros

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

How many complex zeros, counted with multiplicity, does $p(x) = x^6 - 2x^3 + 5$ have?

(A) 3
(B) 5
(C) 6
(D) 2

Question 2

NO CALCULATOR

A polynomial with real coefficients has zeros 2 and $1 + 3i$. Which must also be a zero?

(A) -2
(B) $3 + i$
(C) $-1 - 3i$
(D) $1 - 3i$

Question 3

NO CALCULATOR

What are the zeros of $p(x) = x^3 + 4x$?

(A) $0,\ 2,\ -2$
(B) $0,\ 2i,\ -2i$
(C) $0,\ 4i$
(D) 0 only

Question 4

NO CALCULATOR

A polynomial of degree 5 with real coefficients has exactly two non-real zeros. How many real zeros (counted with multiplicity) does it have?

(A) 1
(B) 2
(C) 3
(D) 5

Question 5 (Free response) NO CALCULATOR

Find all complex zeros: (a) $x^2 + 9$ (b) $x^4 - 16$ (c) $x^3 - x^2 + 4x - 4$ (hint: group).